

Ejemplo 1)

$\Phi = 4\Phi_{ABCD}$ $a = 1\text{m}$

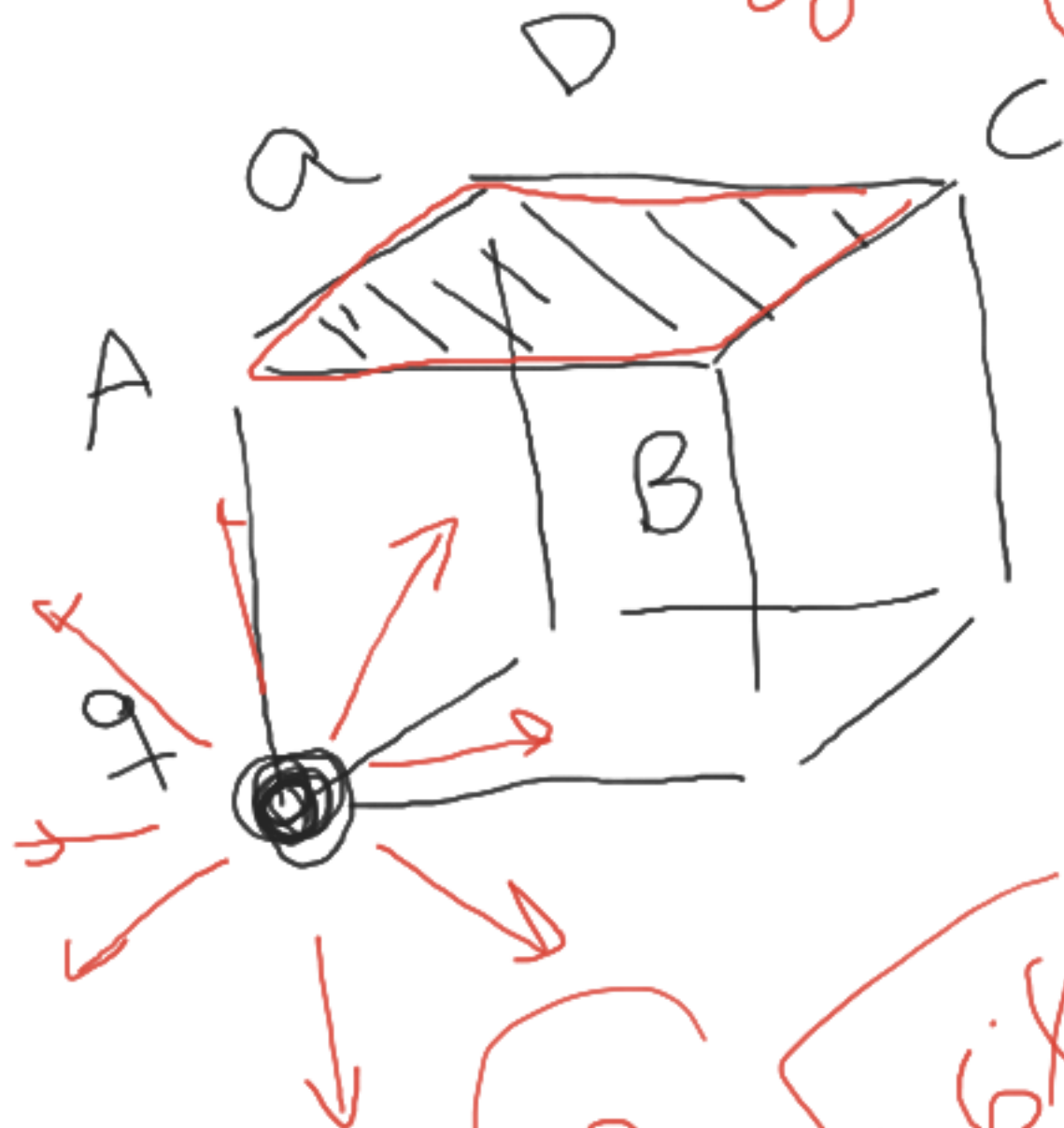
$\frac{q}{\epsilon_0}$



$6\left(\frac{q}{\epsilon_0}\right)$

$\frac{q}{\epsilon_0}$
 $\frac{1}{2}a$

$\frac{q}{24\epsilon_0}$



$$\Phi = \oint \vec{E} \cdot d\vec{S} = \frac{q}{\epsilon_0}$$

$$= 11 E \cdot S_1$$

$\frac{q}{\epsilon_0}$



$$g_p = -\vec{\nabla} \cdot \vec{p} = -\frac{\partial p}{\partial x}$$

$$\vec{p} = \underbrace{(ax^2 + b)}_P \hat{x}$$

$$g_p = -\frac{\partial (ax^2 + b)}{\partial x}$$

$$g_p = -2ax$$

b) Q.T. polarización

$$Q_{TP} = \int \sigma_p dS + \int p_v dV$$

$$\sigma_p \left(\frac{1}{2} \right) \overline{p_p} \left(\frac{1}{2} \right) \sigma_{p2}$$

$$\int \sigma_P dS = \int \sigma_{P1} dS_1 + \int \sigma_{P2} dS_2$$

$$\sigma_{P1} \quad \sigma_{P2} = \sigma_{P1} \int dS_1 + \sigma_{P2} \int dS_2$$



$$\sigma_{P1} = -b$$

$$\sigma_{P2} = aL^2 + b$$

$$\sigma_v = \frac{2ax}{L^2}$$

$$\int \sigma_P dS = (-b)A + (aL^2 + b)A$$

$$\int \sigma_P dV = \int_0^L -2ax A dx = -2aA \int_0^L x dx$$

$$dV = A dx \quad \Rightarrow \quad -2aA \left(\frac{x^2}{2} \right)_0^L = -aAL^2$$

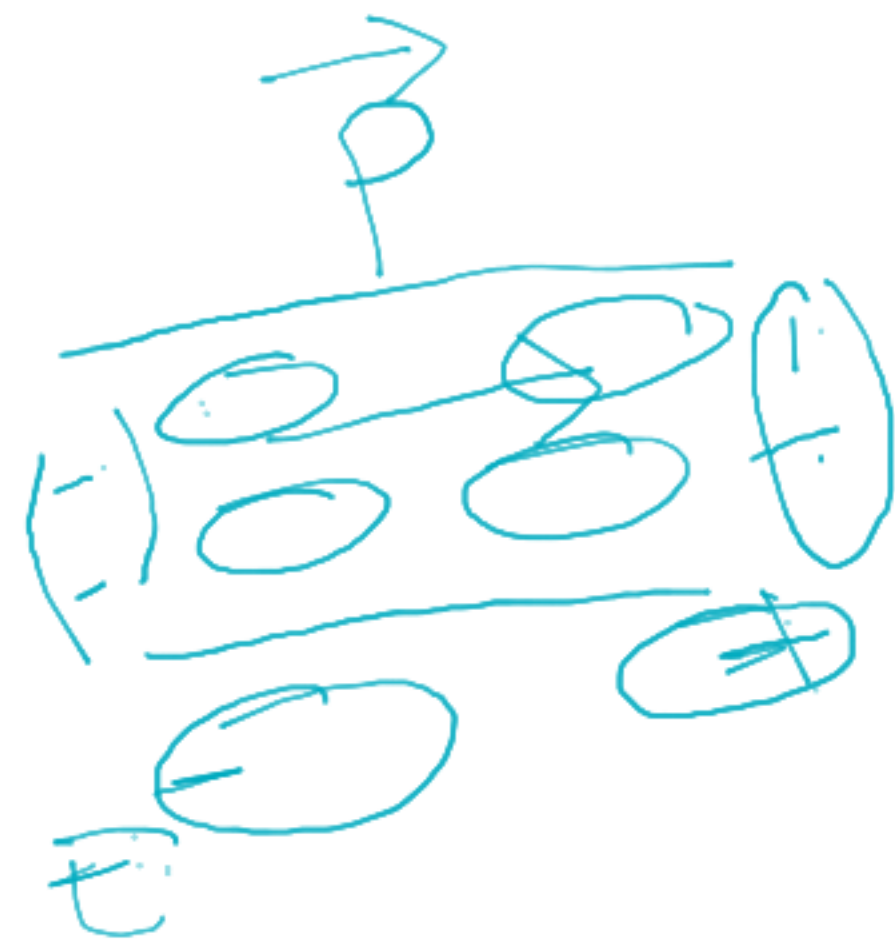
$$Q_{HP} = \int \sigma_p ds + \int p dv$$

$$= -bA + (aL^2 + b)A + (-aAL^2)$$

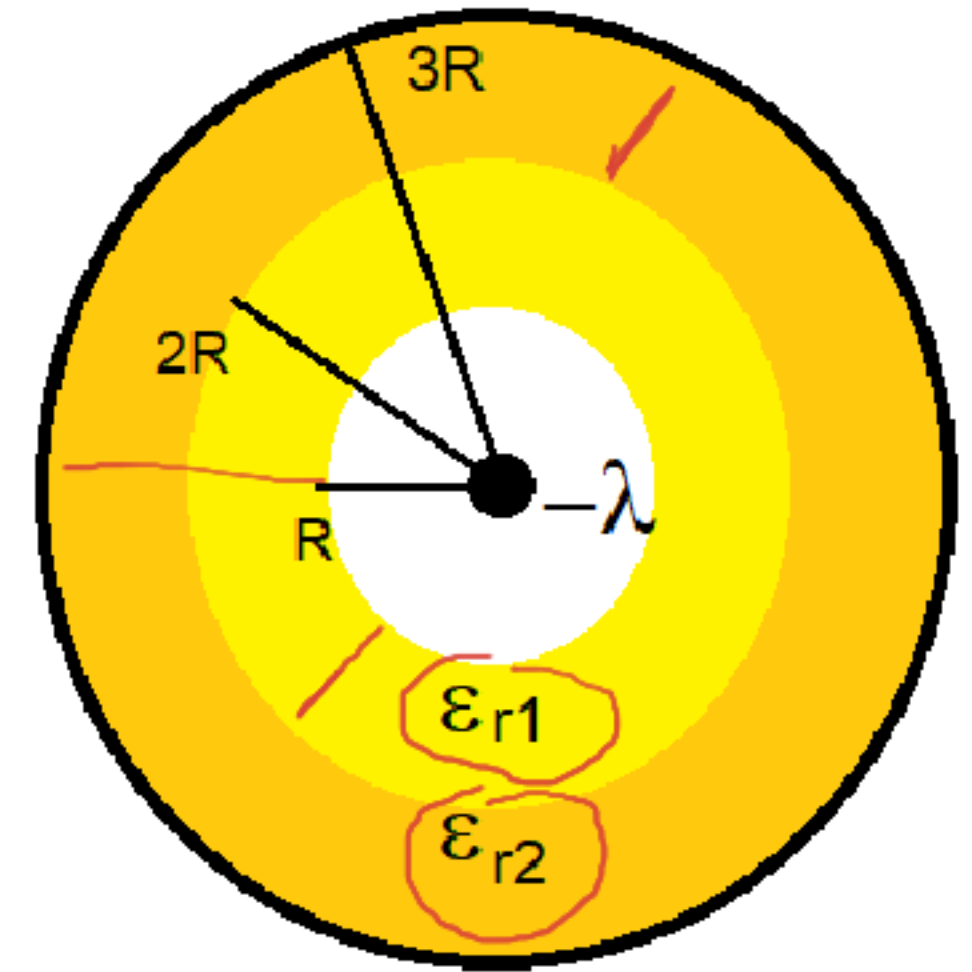
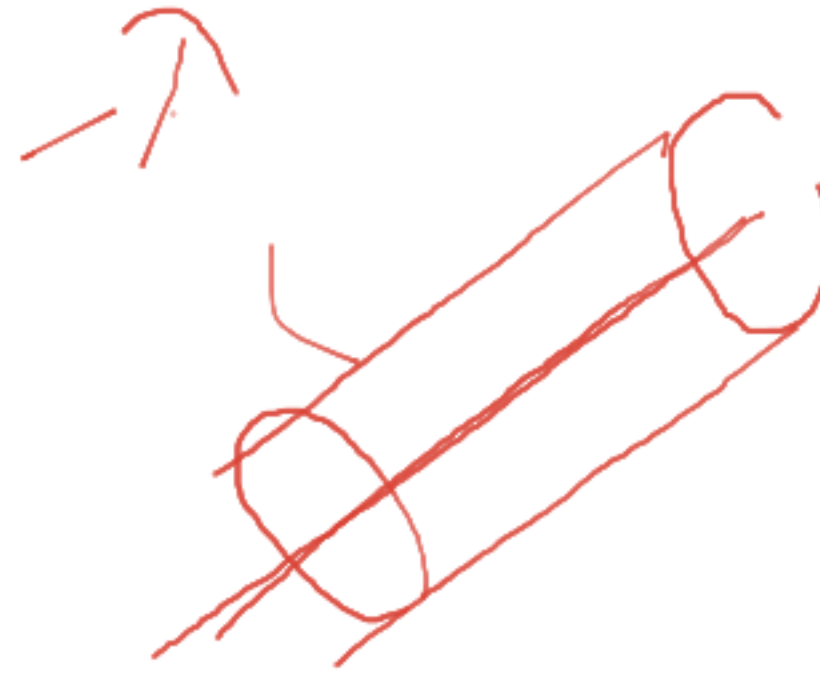
$$Q_{HP} = -bA + aL^2A + bA - aAL^2$$

$$Q_{TP} = 0$$

$$(aL^2A)$$



La figura muestra la sección transversal de un hilo conductor muy largo con densidad lineal de carga $-\lambda$ negativa que está rodeada de una zona de aire de radio R alrededor de la cual están situadas 2 capas cilíndricas concéntricas.



La primera de ellas de radio interior R y exterior $2R$, tiene permitividad relativa $\epsilon_{r1} = 2$, la segunda de radio interior $2R$ y radio exterior $3R$ tiene permitividad relativa $\epsilon_{r2} = 3$. Calcular

- El campo eléctrico en todo el espacio.
- Las densidades de carga de polarización.
- La diferencia de potencial entre un punto a distancia R del hilo y otro a distancia $3R$

$$\epsilon_r = \frac{\epsilon}{\epsilon_0}$$



$$\text{div } \vec{B} = \rho$$

$$\int \text{div } \vec{D} dv = \oint \vec{D} \cdot d\vec{S}$$

$$\oint \vec{D} \cdot d\vec{S} = \int \rho dv = Q$$

$$D \oint dS = \int \lambda dl$$

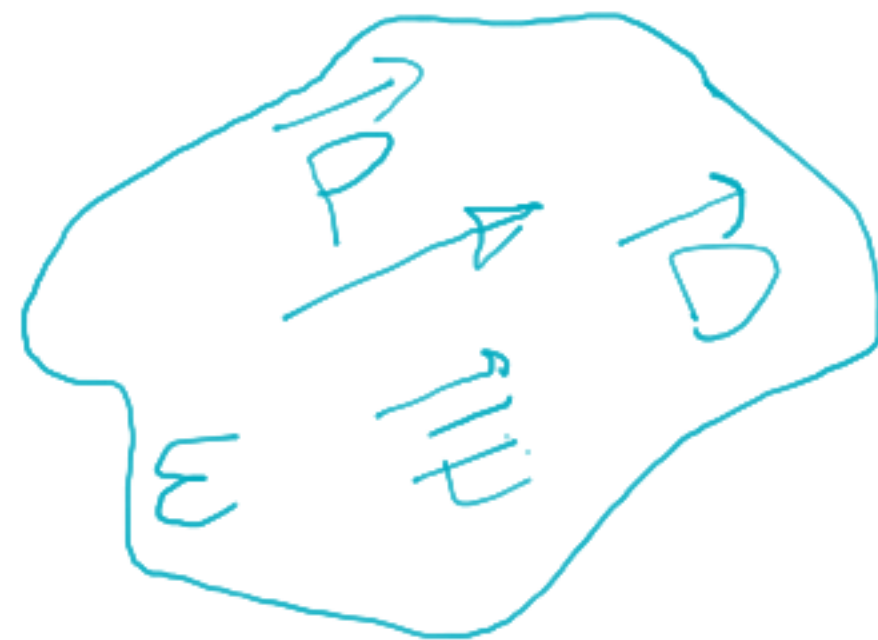
$$D 2\pi r l = \int \lambda dl$$

$$\int \vec{E} \cdot d\vec{S} = \frac{Q}{\epsilon}$$



$$\epsilon_{\text{air}} \epsilon_0$$

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P}$$

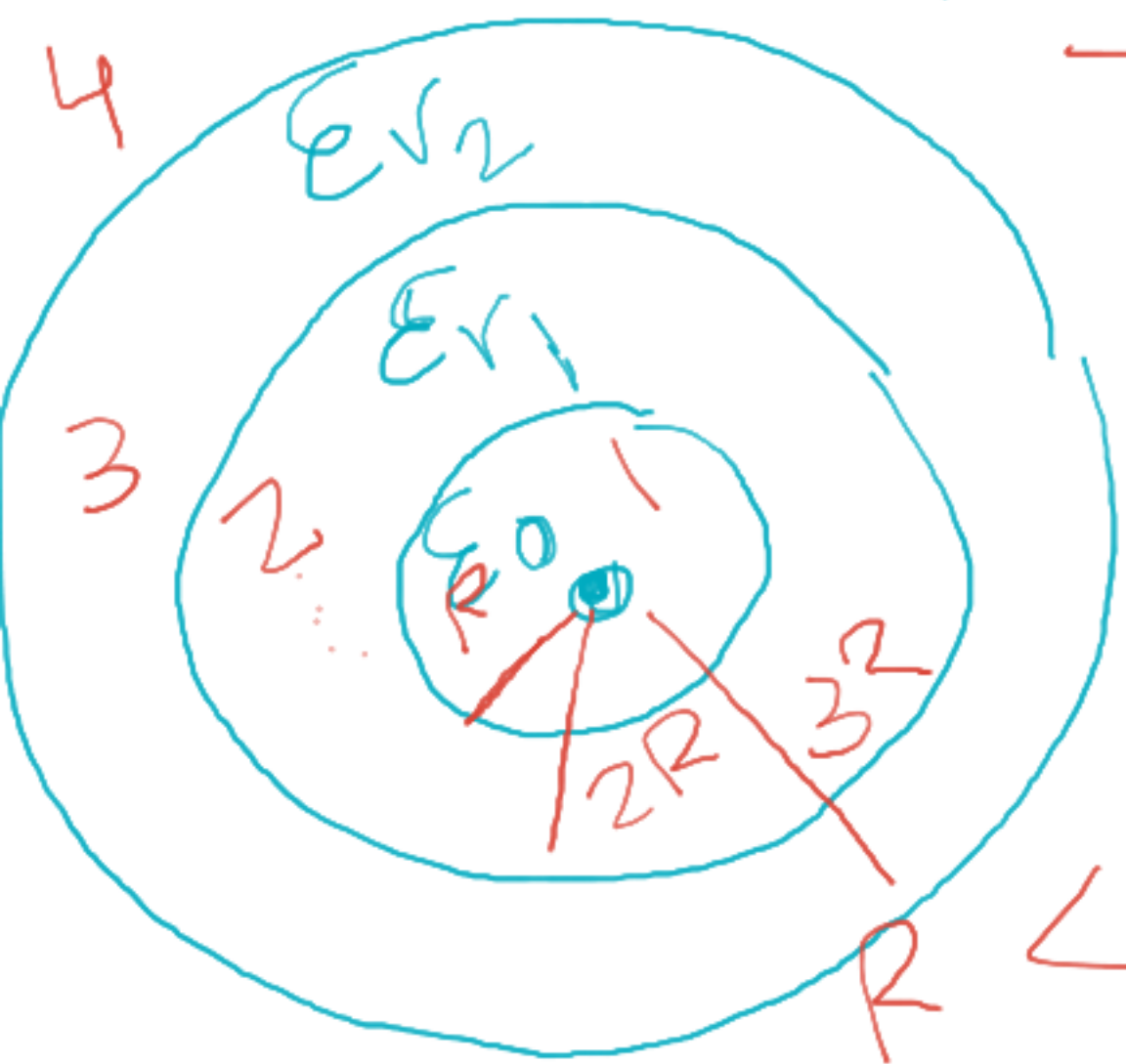


$$\vec{D} = \epsilon \vec{E}$$

$$r < R$$

$$\vec{D} = \epsilon_0 \vec{E}$$

$$\vec{E} = \frac{\vec{D}}{2\pi\epsilon_0 r}$$



$$\epsilon_0 \Rightarrow \vec{E} = \frac{\vec{D}}{\epsilon_0}$$

$$r < R < 2R$$

$$\epsilon_1 = \epsilon_0 \epsilon_{r1}$$

$$\vec{E} = \frac{\vec{D}}{\epsilon_0 \epsilon_{r1}}$$

$$R < r < 2R$$

$$\vec{E} = -\frac{\lambda}{2\pi r} \hat{r} \quad \hat{r}$$

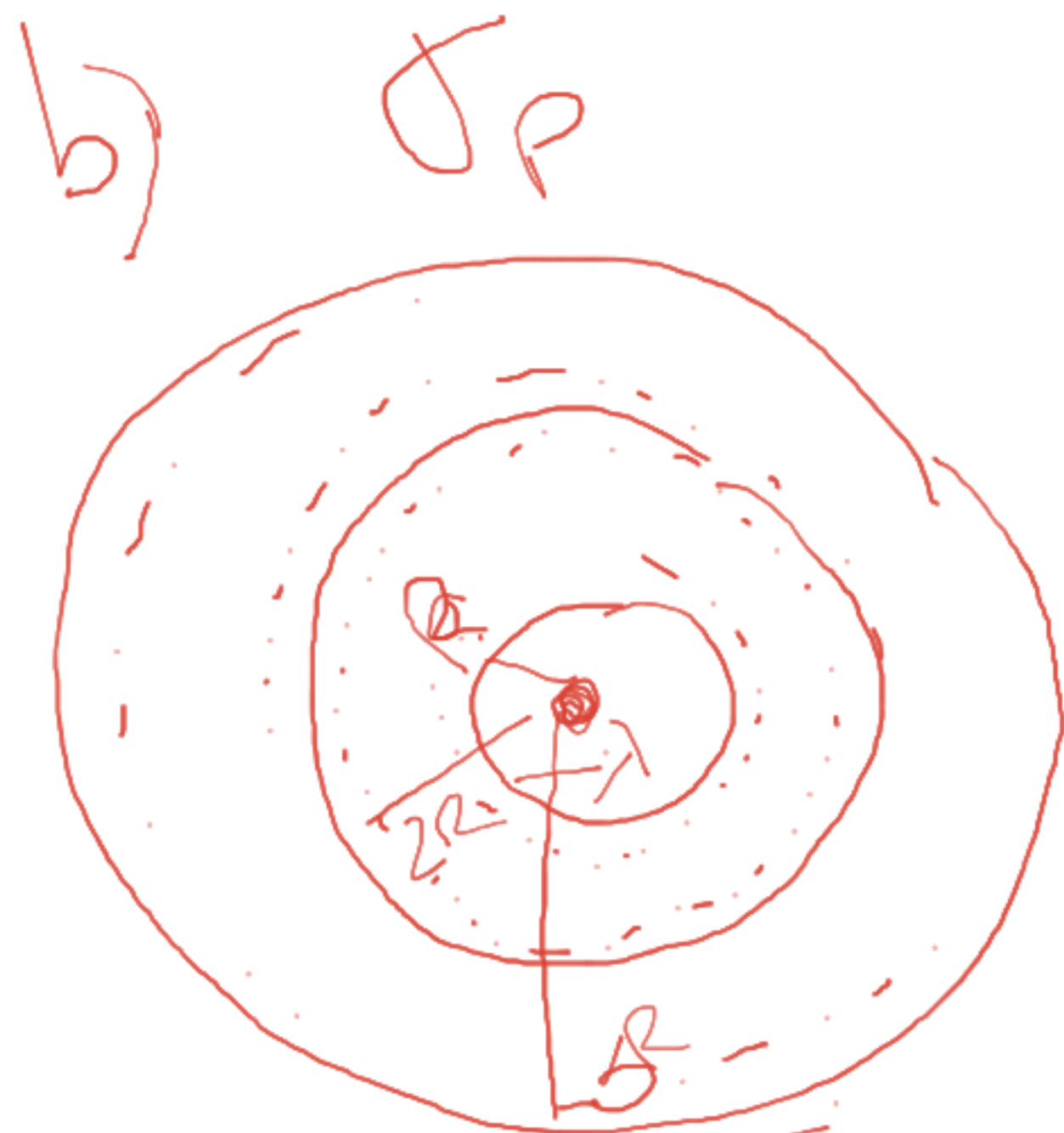
$$\vec{E} = -\frac{\lambda}{4\pi\epsilon_0 r} \hat{r} \quad \hat{r}$$

$$\vec{E} = \frac{\vec{D}}{\epsilon_0 \epsilon_r} = \frac{\vec{D}}{3\epsilon_0}$$

$$2R < r < 3R$$

$$\vec{E} = -\frac{\lambda}{6\pi\epsilon_0 r} \hat{r} \quad \hat{r}$$

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P} \quad \checkmark$$



$$\vec{D} = \epsilon \vec{E}$$

$$\sigma_p = \vec{P} \cdot \hat{n}$$

$$\vec{P} = \vec{D} - \epsilon_0 \vec{E}$$

$$\sigma_p (r=R) = \vec{D} - \epsilon_0 \frac{D}{\epsilon_1}$$

$$\vec{P} = \vec{D}$$

$$\left(1 - \frac{\epsilon_0}{\epsilon_1 \epsilon_3} \right)$$

$$\vec{P} = \vec{D} \left(1 - \frac{1}{\epsilon_r} \right) = \left(\epsilon_r - 1 \right) \frac{1}{\epsilon_r} \vec{D}$$

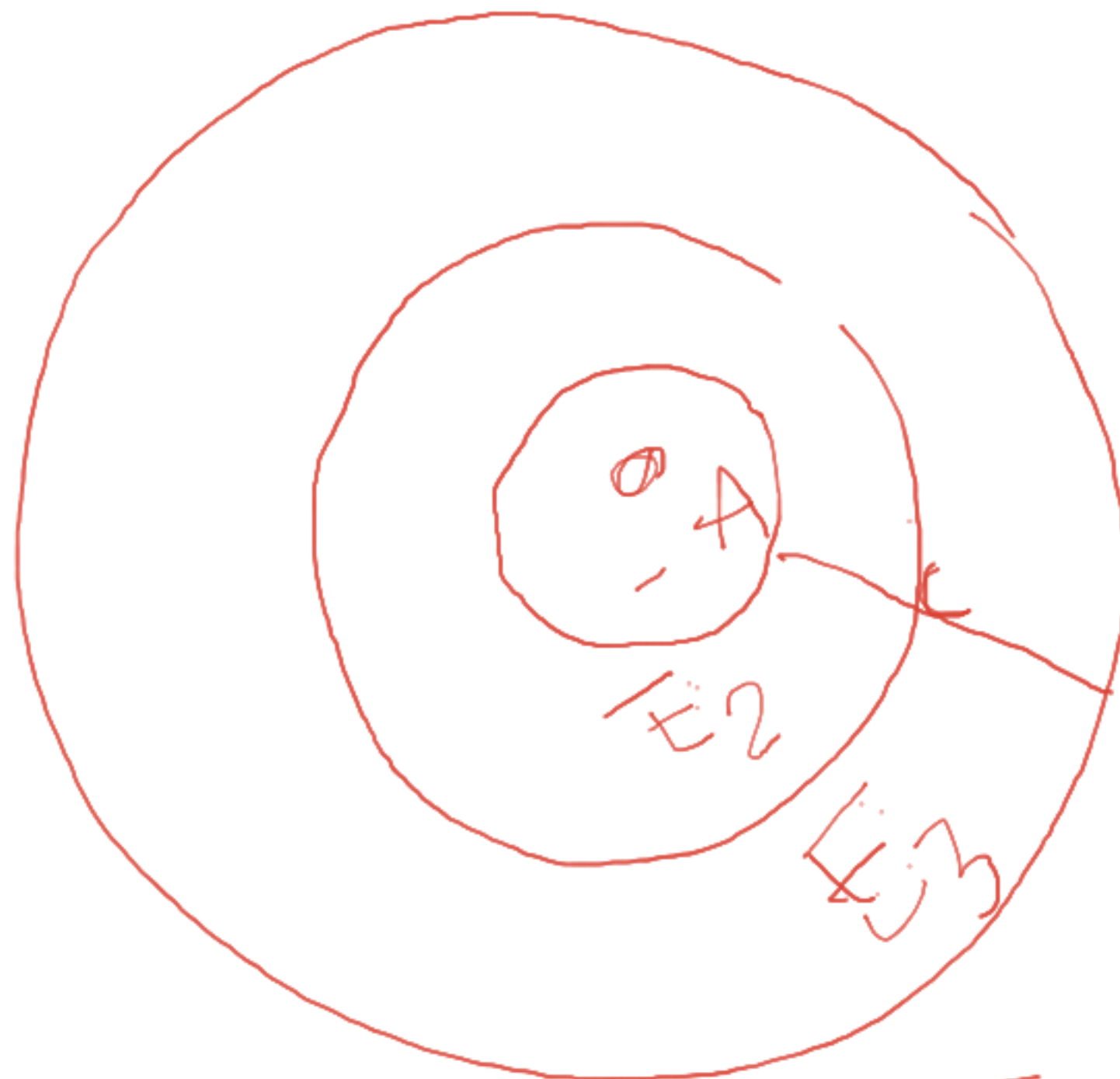
$$\vec{P} = \frac{1}{2} \left(\frac{-\lambda}{2\pi r} \right) \hat{u}_r$$

$$\vec{P}(r=R)$$

$$\oint \vec{P}_1(r=R) = \frac{-\lambda}{4\pi R}$$

$$\oint \vec{P}_1 = \frac{\lambda}{4\pi R}$$





$$V_A - V_B = \int_A^B E_{\text{all}} = \int_A^C E_2 + \int_C^B E_3$$